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NAAC GRADE A+

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Subject Name: TECHINCAL SKILLS **Subject Code:**

WORKSHEET 8

AIM: To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

S/W Requirement: Oracle Database Express Edition and pgAdmin

OBJECTIVES:

1. To understand the concept and usage of stored procedures in database systems.
2. To implement stored procedures for performing CRUD operations efficiently.
3. To enhance data security by restricting direct access through stored procedures.
4. To reduce code redundancy by using reusable database procedures.
5. To improve performance of database operations using precompiled stored procedures.

Practical / Experiment Steps

Tables Creation & Insertion

	emp_id [PK] integer	emp_name character varying (50)	manager_id integer	department character varying (50)	salary integer
1	2	Ravi	1	Engineering	80000
2	3	Neha	1	Engineering	82000
3	4	Karan	2	Engineering	60000
4	5	Simran	2	Engineering	62000
5	6	Pooja	3	Engineering	61000
6	7	Rahul	3	Engineering	64000
7	8	Arjun	1	HR	70000
8	1	Amit	[null]	Management	120500

Step 1. Create or replace a procedure for Updating Employee Salaries

```
create or replace procedure update_salary_proc(IN P_EMP_ID INT,INOUT P_SALARY
NUMERIC(20,3),OUT STATUS VARCHAR(20))
AS
$$
DECLARE
CURR_SAL NUMERIC(20,3);
```

```
BEGIN

SELECT SALARY+P_SALARY INTO CURR_SAL FROM employees WHERE EMP_ID=P_EMP_ID;
IF NOT FOUND THEN
RAISE EXCEPTION 'EMPLOYEE NOT FOUND';
END IF;

UPDATE employees
set salary= curr_sal
where emp_id=p_emp_id;

P_salary:=curr_sal;
status:='SUCCESS';

EXCEPTION
WHEN OTHERS THEN
IF SQLERRM LIKE '%EMPLOYEE NOT FOUND%' THEN
STATUS:= 'EMPLOYEE NOT FOUND';
END IF;

END;

$$ LANGUAGE PLPGSQL;
```

Step 2 Calling of Stored Procedure

```
DO
$$
DECLARE
EMP_ID INT:=1;
STATUS VARCHAR(20);
SALARY NUMERIC(20,3):=500;
BEGIN
CALL update_salary_proc(Emp_id,salary,status);
RAISE NOTICE 'YOUR STATUS IS % AND YOUR UPDATED SALARY IS  %',STATUS,salary;
END;
$$
```

Output:

	emp_id [PK] integer	emp_name character varying (50)	manager_id integer	department character varying (50)	salary integer
1	2	Ravi	1	Engineering	80000
2	3	Neha	1	Engineering	82000
3	4	Karan	2	Engineering	60000
4	5	Simran	2	Engineering	62000
5	6	Pooja	3	Engineering	61000
6	7	Rahul	3	Engineering	64000
7	8	Arjun	1	HR	70000
8	1	Amit	[null]	Management	121000

```
NOTICE:  YOUR STATUS IS SUCCESS AND YOUR UPDATED SALARY IS  121000.000
DO

Query returned successfully in 85 msec.
```

Outcomes:

1. To demonstrate the ability to create and execute stored procedures for various database operations.
2. To apply stored procedures for performing secure and efficient CRUD operations.
3. To analyze the impact of stored procedures on database performance and execution efficiency.
4. To design modular and reusable database logic using stored procedures.